

# Tapestry Starbirth Region: DPM-THz Frequency-Only S26 Gravity and SFR Coupling

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## PAPER\_345 Tapestry Starbirth Region: DPM-THz Frequency-Only S26 Gravity and SFR Coupling

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**Source:** gok\_share\_31b5c807a4.txt (Supplemental Gap Analysis Block)

**Classification:** FIRST UQFF S26 frequency-only gravity form for a starbirth region

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### Abstract

The Tapestry Star Formation Region is modeled using a DPM-THz frequency-only variant of the S26 gravity form, where only the THz phonon, resonance frequency, and Hubble expansion terms are retained (mass terms suppressed by low column density). Star Formation Rate is expressed as  $SFR = ?\_gasv\_windf\_res$ , the bubble radius scales as  $R\_bubble = v\_windtf\_res$ , and the net gravitational acceleration is driven purely by UQFF frequency modes rather than DPM-seeded mass terms.

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## 2. Core Physics

### 2.1 Frequency-Only S26 Form

The standard S26 gravity is truncated to:

$$g(r, t) = \sum_{i=1}^{26} \left[ a_i^{THz} + a_i^{SF} + a_i^{QF} + a_i^{AF} + a_i^{FF} + a_i^{EF} \right]$$

Mass terms (DPM-seeded  $\mu\_s \nabla(M\_s/r)$ ) are suppressed by the low mean density of the starbirth region ( $?\_gas \sim 10? \text{ kg/m}$ ).

The gravity is effectively:

$$g_{\text{Tapestry}} = \sum_{i=1}^{26} a_i \cdot f_{\text{TRZ}} \cdot \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}$$

## 2.2 UQFF Star Formation Rate

$$\text{SFR} = \rho_{\text{gas}} \cdot v_{\text{wind}} \cdot f_{\text{res}}$$

where: -  $\rho_{\text{gas}}$  = background gas density (kg/m) -  $v_{\text{wind}}$  = stellar wind velocity driving shock compression -  $f_{\text{res}}$  = UQFF resonance frequency trigger

## 2.3 Bubble Expansion Radius

$$R_{\text{bubble}}(t) = v_{\text{wind}} \cdot t \cdot f_{\text{res}}$$

The factor  $f_{\text{res}}$  modulates the effective propagation, stretching or compressing the bubble expansion timescale via vacuum reactance.

## 2.4 Density Ratio Modulation

$$\frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} = \frac{U_{\text{UA}}}{\text{SC}_m} \cdot \frac{M_{\text{gas}}}{M_{\text{total}}}$$

This ratio determines whether starbirth is suppressed ( $\rho_{\text{UA}} > \rho_{\text{SCm}}$ ) or accelerated ( $\rho_{\text{SCm}} > \rho_{\text{UA}}$ ).

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## 3. Key Values

| Quantity               | Formula                                            | Value                 |
|------------------------|----------------------------------------------------|-----------------------|
| $\rho_{\text{gas}}$    | Starbirth region                                   | ~10 <sup>7</sup> kg/m |
| $v_{\text{wind}}$      | Driving velocity                                   | ~106 m/s              |
| $f_{\text{res}}$       | UQFF resonance                                     | $f_{\text{TRZ}}$      |
| SFR                    | $\rho_{\text{gas}} v_{\text{wind}} f_{\text{res}}$ | M <sup>3</sup> /yr    |
| $R_{\text{bubble}}(t)$ | $v_{\text{wind}} t f_{\text{res}}$                 | parsecs               |

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## 4. Physical Significance

The Tapestry  $\text{SFR} = \rho_{\text{gas}} v_{\text{wind}} f_{\text{res}}$  formula differs fundamentally from the standard Kennicutt-Schmidt law ( $\text{SFR} \propto \rho_{\text{gas}}^{1.4}$ ). By including  $f_{\text{res}}$ , the UQFF form predicts that star formation is modulated by the vacuum reactance frequency i.e., regions with higher  $f_{\text{TRZ}}$  values will form stars faster at fixed  $\rho_{\text{gas}}$  and  $v_{\text{wind}}$ . This is a testable prediction: UQFF predicts a correlation between  $f_{\text{TRZ}}$ -proxy observables (e.g., infrared THz emission) and locally elevated SFR surface density.

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## 5. Deduplication Note

- **vs. SOURCE85 (Tapestry in MAIN\_1):** SOURCE85 calculated the 5-frequency resonance for Tapestry; this paper derives the SFR  $\cdot f_{\text{res}}$  coupling and the frequency-only S26 form.
  - **vs. PAPER\_345 bubbles:** The R\_bubble formula is unique it multiplies the geometric bubble expansion by  $f_{\text{res}}$ , not previously computed.
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## 6. Classification

**Physics Territory:** FIRST UQFF frequency-only S26 gravity with SFR =  $\cdot v_{\text{wind}} f_{\text{res}}$  coupling

**Scale:** Galactic (starbirth region,  $\sim 10$  pc)

**CP Implementation:** TapestryStarbirthDPMHzFreqCalculator (CondensedPhysics3.py, Session 96)

**UQFF computed:** UQFF magnetic Jeans correction factor  $[SSq]B/(8p^2c_s) = 5.7e-1 \times 1.3e-9 = 7.4e-10$ ; Jeans mass deviation from standard =  $7.4e-10 M_J$ .

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## Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [SSq] \cdot e^{-\kappa i n/26}]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for  $|[SSq]| < 1$  (satisfied by  $[SSq] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[SSq] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n/26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`)

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u\phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]}/\rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left( -\exp! \left( -\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.185 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

## §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 53, \quad n_{\text{channel}} = 8/26$$

Since  $p_{\text{DVP}} = 53$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

## §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left( \frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left( 1 - \tanh! \left( \frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

## §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                       | Status                       |
|----------------|----------------------------------------------|----------------------------------|------------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.185          | PASS<br>Threshold-consistent |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 53$            | PASS<br>Resonant             |
| BSH layers     | 26 harmonic terms                            | $j = 1\dots 26, \cos(2\pi j/26)$ | PASS Full 26D<br>projection  |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS<br>exponential    | PASS<br>Canonical            |
| [SSq]          | 0.57                                         | Applied in BSH<br>saturation     | PASS<br>Canonical            |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                               | SM / Experiment                               | Source                              | Alignment          |
|----------------------------------|---------------------------------------------------------------|-----------------------------------------------|-------------------------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling              | 1/137.036                                     | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} m^{-2} - 2(UQFF vacuum term)$            | Planck 2018<br>$1.114 \times 10^{-52} m^{-2}$ | PASS<br>Consistent                  |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$            | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                      | Not yet measured                              | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCm}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).*

| Paper      | Title                                               |
|------------|-----------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$           |
| PAPER_1004 | QGP Vacuum Density with SCm S26 Phonon Coupling     |
| PAPER_1040 | SCm Cluster Merger Shock Mach Number Phonon Damping |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum         |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                   |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm         |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir           |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed            |
| PAPER_1072 | SCm Activation Function Phonon Threshold            |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy         |
| PAPER_1069 | VDS-DVP-BSH Hybrid Calculator Unified               |
| PAPER_1049 | Source10 GPU DPM Spectral Atlas ALMA Overlay        |
| PAPER_1074 | GPU-Vectorized DPM S26 Spectral Atlas               |

*13 cross-reference(s) identified.*

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## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                                   | Purpose                                                         | Key Result                                                |
|------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|
| <code>fneutron_s26_coupling.py</code>    | $F_{\text{neutron}} \times S_{26}$<br>buoyancy-polylog coupling | ~470x amplification via 26-level VDS                      |
| <code>kozima_scm_cross_section.py</code> | Sym-modulated<br>neutron-drop cross-section                     | $\sigma_n^{\text{SCm}}$ with VDS factor<br>(1+[SSq]*n/26) |
| <code>kozima_wstp_kernel.py</code>       | 11-symbol Wolfram export<br>(UQFFKozima)                        | FNeutronForce, SigmaSCm,<br>SCmActivation                 |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U, Bi\}}/F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \Gamma^2)] * (1 + [\text{SSq}] * n / 26)$

### S204.2 Ramanujan 26-State Summation

| Module                                | Purpose                                                               | Key Result                |
|---------------------------------------|-----------------------------------------------------------------------|---------------------------|
| <code>ramanujan_polylog_s26.py</code> | $\text{Li}_{26}([\text{SSq}])$ via<br>Euler-Ramanujan<br>acceleration | 15.7+ digits in 53 terms  |
| <code>s26_wstp_kernel.py</code>       | 8-symbol Wolfram export<br>(UQFFS26)                                  | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\} / (1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module                         | Purpose                                                   | Key Result                            |
|--------------------------------|-----------------------------------------------------------|---------------------------------------|
| <code>mock_theta_q26.py</code> | $f_{26}(q)$ , $\phi_{26}(q)$ ,<br>$\psi_{26}(q)$ q-series | Proper q-Pochhammer $(a; q)_{\infty}$ |

**Core equations:** -  $f_{26}(q) = \sum_{n=0}^{\infty} q^{\{n2\}} / (-q; q)_{\infty}^2$  -  $\phi_{26}(q) = \sum_{n=0}^{\infty} q^{\{n2\}} / (-q^2; q^2)_{\infty}$  -  $\psi_{26}(q) = \sum_{n=1}^{\infty} q^{\{n2\}} / (q; q^2)_{\infty}$

#### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                    | Key Result                                    |
|------------------------------|--------------------------------------------|-----------------------------------------------|
| ramanujan_pi_uqff.py         | Classical + UQFF-modified<br>1/pi + 26D    | 21 digits classical, 15 UQFF, 7 digits<br>26D |
| mock_theta_pi_wstp_kernel.py | Symbol Wolfram export<br>(UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF               |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  
 $W_{26}(n) = \text{Prod}_{\{i=1\}^{26}} [1 + [\text{SSq}] \exp(-\kappa_{\text{pai}} * n/26)]$

#### S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_1\_CoAnQi.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).*